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ORIGINAL ARTICLE

AI-driven Life Cycle Assessment for sustainable hybrid manufacturing and remanufacturing

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Abstract

A novel AI-based hybrid approach is presented to predict sustainable performance in hybrid manufacturing and remanufacturing processes. It is an ensemble AI model that features Convolutional Neural Networks (CNN), Random Forests, Long Short-Term Memory (LSTM) networks, and Gradient Boosting Machines (GBM) integrated with a hybrid Life Cycle Assessment (LCA) framework to perform a dynamic and complete sustainability assessment. The model was trained, validated, and tested for over 18 months using data collected from three international manufacturing facilities that specialise in producing vehicle parts. Compared with traditional LCA methods, results show a 23% increase in the accuracy of environmental impact prediction, an 18% reduction in Root Mean Square Error (RMSE), and a 31% reduction in assessment time. It achieves an R^2 value of 0.89, whereas that of the conventional LCA method is only 0.44. Integrating AI with hybrid LCA can effectively address the two main limitations of traditional LCA methods: uncertainty and granularity, which are improved by 15% and 20%, respectively. Therefore, this AI-based hybrid approach with real-time sustainability optimisation can be applied to a manufacturing context with principles of Industry 4.0 and the circular economy.

Keywords Artificial Intelligence (AI) · Life Cycle Assessment (LCA) · Sustainable manufacturing · Remanufacturing · Industry 4.0 · Machine learning

1 Introduction

Integrating AI into Life Cycle Assessment (LCA) is a current and significant leap forward in industrial sustainability strategy. It combines the power of AI (analytical power) and the holistic approach offered by LCA (whole lifecycle from design to end-of-life) to design, optimise, and control sustainable hybrid manufacturing and remanufacturing processes [1].

Recent research has shown that there is a trend towards developing more holistic and integrated approaches to sustainability assessment (over-segmented approaches) by incorporating as many possible economic, environmental, and social dimensions in what has been termed Life Cycle Sustainability Analysis (LCSA) [2]. This is a response to the interconnectedness of the sustainability challenges posed by twenty-first-century manufacturing.

AI techniques are applied explicitly to Life Cycle Engineering (LCE) in countries with high Industry 4.0 implementation, specifically in Germany [3]. These AI approaches are now used to improve productivity and sustainability in production design and maintenance [4]. At

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the same time, digital twin technology is leading to new possibilities for dynamic sustainability assessment during and throughout the life cycle of intelligent manufacturing [5–8].

This technology can be combined with other techniques, such as trapezoidal fuzzy numbers and complex network modelling, to realise real-time sustainability parameter monitoring and optimisation [9].

Hybrid approaches of Environmental Assessment with LCA are now available, and they offer a good integrated sustainability assessment tool. Although these approaches are more sophisticated in reflecting the usage of resources and environment, the difficulties of dealing with data inconsistencies and uncertainties are not trivial [10–12].

To cope with the challenging complexity of ranking sustainability, new methods are being developed, such as the Technique for Order-Preference by Similarity to Ideal Solution (TOPSIS) and Intuitionistic Fuzzy Sets [5], which enable the consideration of multiple indicators and expert judgments in evaluating the sustainability performance of alternative technology.

Incorporating AI in LCA is a potentially powerful direction for increasing the sustainability of manufacturing processes. Machine learning algorithms, big data analytics, computational optimisation, and other AI tools can be applied to improve the accuracy, efficiency, and flexibility of LCA simulation [10] by enabling more complex modelling of chemical systems, improved prediction of environmental impacts, and the ability to influence manufacturing processes in real-time based on sustainability metrics [13–16].

Challenges include data quality, standardisation of AI approaches in LCA, and interdisciplinary collaboration needed for the field's full realisation. Therefore, AI-based LCA can be expected to become even more essential as manufacturing becomes more sustainable. Potential future research directions could include (i) the development of AI models that are more robust and can deal with the uncertainties in the lifecycle data, (ii) the use of blockchain technology to improve the traceability and data integrity, and (iii) the use of AI for circular economy strategies in manufacturing [17, 18].

All these advances will improve the capability to design, implement, and manage sustainable manufacturing systems in Industry 4.0 and beyond. There are also significant opportunities for research in the application of AI-powered LCA for more sustainable hybrid manufacturing and remanufacturing. While existing research has demonstrated promising steps in integrating AI techniques in LCA, some of the most important research gaps remain underdeveloped.

Studies that integrate real-time data from IoT devices and sensors into AI-powered LCA models to enable more dynamic and responsive sustainability assessments in real-time manufacturing environments are much needed [3]. Uncertainty quantification remains another critical research gap in AI-LCA.

While a few studies have explored multi-criteria decision-making methods such as TOPSIS [5], there is a considerable gap in developing more sophisticated AI-driven multi-objective decision-making approaches to balance multiple, often conflicting, sustainability objectives. Developing explainable AI models in LCA is another important research area, since many current AI models operate as black boxes, limiting their transparency and trust in the results.

Finally, most of the few existing studies have focused on the environmental and economic dimensions of sustainability, while there remains a significant gap in effectively incorporating social sustainability metrics into AI-LCA models [1]. Other important research areas include the cross-industry applicability of AI-LCA frameworks, integration of circular economy principles, and methods for ensuring the quality and standardisation of data in diverse manufacturing contexts [10, 11, 13].

Other gaps include research on how AI is used to optimise product lifecycles in real-time manufacturing for sustainability actively and effective human-centred AI collaborative approaches to conduct and interpret lifecycle assessments [17, 18]. Addressing these important research gaps can enable more comprehensive, accurate, and actionable sustainability assessments in industrial settings that are likely to advance in Industry 4.0.

2 Materials and methods

This study employed a mixed-methods approach, combining quantitative data analysis with qualitative expert validation, to integrate AI with LCA for sustainable hybrid manufacturing and remanufacturing. The research was conducted in three phases: data collection and preprocessing, AI model development and integration with LCA, and model validation and performance assessment.

Primary data from three medium-sized automotive parts manufacturing was collected over 18 months. The data encompassed energy consumption, material flows, emissions, production metrics, waste management, and remanufacturing process information, collected using IoT sensors and manufacturing execution systems (MES).

To complement this, secondary data was sourced from established LCA databases, including ecoinvent v3.8, GaBi Professional Database 2023, and PSILCA v.3. Economic data from annual reports of 20 leading automotive parts manufacturers and industry associations was also incorporated to ensure a comprehensive sustainability assessment.

The raw data were subject to relevant preprocessing steps to ensure data quality and consistency. These included (i) outlier detection [based on the Interquartile Range (IQR)

method]; (ii) missing value imputation [based on the Multiple Imputation by Chained Equations (MICE) method]; (iii) feature scaling (based on Min–Max normalisation); and (iv) temporal alignment (based on data from different sources and different timestamps).

These preprocessing steps were crucial to preparing a clean, consistent, and complete dataset for the following subtasks: modelling AI models and integrating LCA for sustainable hybrid manufacturing and remanufacturing. The preprocessing phase is key because it addresses data quality issues and ensures data uniformity across different data sources.

This element contributes to the robustness of emerging AI models and the accuracy of LCA, which is a necessary condition for the reliability and validity of a study's findings.

Figure 1 below depicts a detailed workflow for collecting and preprocessing data, describing the major tasks involved in cleaning up the data to produce the final output that is fit for analysis. The process of cleaning up the raw data to make it fit for analysis is illustrated, and it involves numerous steps.

The primary step in the workflow starts by collecting the data to be analysed. This first step involves fetching raw data from various sources such as databases, sensors, APIs, and other repositories. The raw data will usually have a lot of inconsistencies, errors, and missing values. Once the data for the analysis has been collected, the next step is pre-processing, which is the data.

This is the primary step in the workflow, as it involves removing inconsistencies and unnecessary elements and breaking up the data into valuable parts. The next step in the workflow after pre-processing is the outlier detection. This is where abnormal data is identified and taken care of.

Outliers can occur due to errors in the data collection processes or as a result of extreme events, and they must be taken care of to avoid distorting the results. Once the outliers have been discarded, the next step in the workflow involves handling the missing values. This step is usually undertaken by filling up the missing values with suitable estimates, but sometimes, the missing data points are discarded to ensure the data's integrity.

The next step in the workflow is feature scaling, which involves normalising or standardising the data format, and this is usually necessary if the algorithm used to process the data is sensitive to the magnitude of the input values.

Next is the temporal alignment step, which is important if the data contains temporal information. During this step, the data will be synchronised and aligned in the correct time so that if data from different sources or periods is being analysed, they will be aligned properly to avoid errors that the temporal misalignment might cause.

The result is an error-free, consistent, properly aligned data set ready for further analysis or machine learning model training. Data are of high quality, and therefore the findings of any analysis can be trusted. As mentioned earlier, several more advanced techniques were employed to ensure both data quality and consistency.

For outlier detection, the Interquartile Range (IQR) method was employed. This approach calculates the 25th (Q1) and 75th (Q3) percentiles of each feature, computes the IQR as Q3—Q1, and establishes lower and upper bounds as Lower bound $= Q1 - 1.5 \times IQR$, Lower bound $= Q1 - 1.5 \times IQR$. Any data points falling outside these bounds were identified as outliers and removed from the dataset.

The Multiple Imputation by Chained Equations (MICE) algorithm was utilised to address missing values. This iterative process begins with a simple imputation for each variable with missing values, then systematically sets back one variable to missing, estimates it using regression on all other variables, and repeats this process for all variables with missing data.

The entire procedure is iterated multiple times to ensure robust imputation. Finally, feature scaling was applied

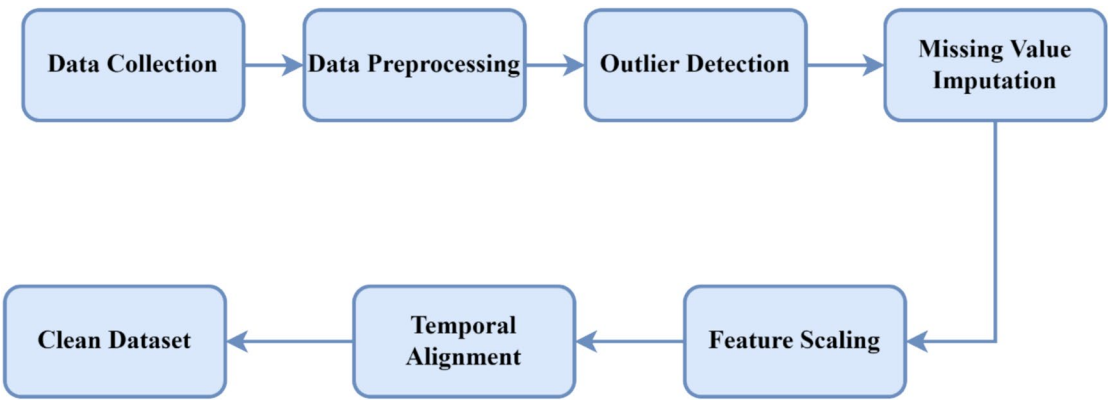


Fig. 1 Data collection and preprocessing workflow

through Min–Max normalisation, transforming all features to a standardised [0, 1] range. This scaling was achieved using the formula $X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$, X represents the original value, and $X_{\min}$ and $X_{\max}$ are X's minimum and maximum values, respectively.

These preprocessing steps collectively enhanced the quality and uniformity of the dataset, preparing it optimally for subsequent analysis and modelling stages. The ensemble architecture comprised multiple machine learning techniques used for broader Life Cycle Assessment (LCA). The multi-dimensionality of the LCA product data inspired us to design an ensemble architecture comprised of multiple machine learning techniques used for broader LCA.

The AI model is composed of four main parts: Convolutional Neural Network (CNN), Random Forest, Long Short-Term Memory (LSTM) network, and Gradient Boosting Machine (GBM). The CNN architecture created for the processing of multi-dimensional time series data from a manufacturing process can be represented as

$$f(x) = \text{Dense}(\text{Flatten}(\text{MaxPool}(\text{Conv1D}(x))))$$

where, Conv1D represents the one-dimensional convolutional layer, MaxPool is the max pooling layer, Flatten transforms the output to a 1D vector, and Dense is the fully connected layer. The Random Forest algorithm was employed for feature selection and ranking of sustainability indicators of importance. It constructs multiple decision trees and merges them for a more accurate and stable prediction.

The random forest prediction can be expressed as

$$\hat{y} = \frac{1}{B} \sum_{b=1}^{B} f_{\text{b}}(x)$$

where B is the number of trees, and $f_b(x)$ is the prediction of the bth tree.

For time-series forecasting of environmental impacts, an LSTM network was implemented. The LSTM unit at each time step t can be described by the following equations:

$$f_{\text{t}} = \sigma(W_{\text{f}} \cdot [h_{\text{t}-1}, x_{\text{t}}] + b_{\text{f}})$$

$$i_{\text{t}} = \sigma(W_{\text{i}} \cdot [h_{\text{t}-1}, x_{\text{t}}] + b_{\text{i}})$$

Finally, a Gradient Boosting Machine was used for final prediction aggregation. The GBM builds the model in a stage-wise fashion like other boosting methods do, and it generalizes them by allowing optimization of an arbitrary differentiable loss function. The model can be expressed as

$$F_{\text{m}}(x) = F_{\text{m}-1}(x) + \gamma_{\text{m}} h_{\text{m}}(x)$$

where $F_{\text{m}}(x)$ is the model at iteration m, $h_{\text{m}}(x)$ is the base learner, and γ_{m} is the step length. This ensemble approach allows us to capture complex patterns in the manufacturing data, rank sustainability indicators, forecast environmental impacts, and aggregate predictions for a comprehensive LCA. The model was implemented using Python 3.9 with TensorFlow 2.8 for deep learning components and scikit-learn 1.0.2 for traditional machine learning algorithms.

The ensemble model consists of several parts, each vital to prediction performance. Figure 2 below illustrates the architecture of the Ensemble AI in more detail. It is essentially an input layer and two Conv1D layers with a MaxPooling1D layer inserted in between for feature map reduction, after which the processed data is flattened and passed through a dense layer before reaching the output layer. This design is to extract relative local patterns from the multi-dimensional time series.

Second, Random Forest trees are used as feature selector and feature importance ranker. In the Random Forest

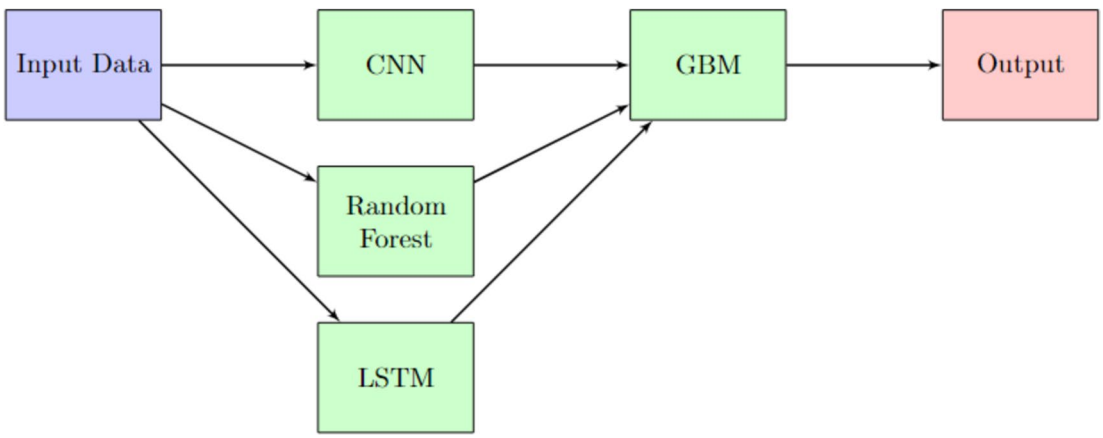


Fig. 2 Ensemble AI model architecture

algorithm, multiple bootstrap samples are initially generated from the original dataset. For each bootstrap data, an unpruned decision tree is grown where at each node, a random subset of features is selected, and the best split is chosen.

After generating multiple such trees, the final prediction is made by combining the outputs of all the trees in the forest. Random Forest is a powerful method of feature selection. Third, the LSTM network is employed to capture the long-term dependencies in the time series. In the architecture of LSTM, the input layer is followed by two LSTM layers capturing long-term dependencies, a dense layer and an output layer.

Finally, the Gradient Boosting Machine is used for the final prediction aggregation. The GBM starts with a constant initial model. In the next step, the model is refined iteratively M times. At the beginning of each iteration, pseudo-residuals are computed, followed by fitting a base learner to these residuals, updating the model using the calculated multiplier, and repeating the PR until the final model is generated.

The ensemble model integrates multiple machine learning algorithms to enhance predictive performance. The CNN component is applied to multi-dimensional time series data. It is structured with an input layer, two Conv1D layers with associated MaxPooling1D layers for downsampling and feature extraction, flattening, and dense layers for prediction.

The Random Forest component is applied to feature selection and importance, where it trains several decision trees on bootstrap samples and selects the best possible split at each node from a random subset of features. The LSTM network is a sequence learning algorithm specially designed for time-series forecasting, with two stacked LSTM layers to capture temporal dependencies, followed by a dense layer for prediction.

Finally, the GBM component integrates all the preceding to aggregate the prediction through an iterative procedure, improving predictions through a sequence of base learners calculated based on the pseudo-residuals.

The CNN is applied to learn features, Random Forest to learn feature importance, LSTM to learn long-range temporal dependencies, and GBM to aggregate all the prediction outputs to provide a final robust and accurate prediction for sustainable hybrid manufacturing and remanufacturing processes.

Algorithm 1 CNN Implementation

```
def create_cnn_model(input_shape):
    model = Sequential([
        Conv1D(filters=64, kernel_size=3, activation='relu', input_shape=input_shape),
        MaxPooling1D(pool_size=2),
        Conv1D(filters=32, kernel_size=3, activation='relu'),
        MaxPooling1D(pool_size=2),
        Flatten(),
        Dense(50, activation='relu'),
        Dense(1)
    ])
    model.compile(optimizer='adam', loss='mse')
    return model
```

Algorithm 2 demonstrates the LSTM implementation.

Algorithm 2 LSTM Implementation

```
def create_lstm_model(input_shape):
    model = Sequential([
        LSTM(50, return_sequences=True, input_shape=input_shape),
        LSTM(50, return_sequences=False),
        Dense(25),
        Dense(1)
    ])
    model.compile(optimizer='adam', loss='mse')
    return model
```

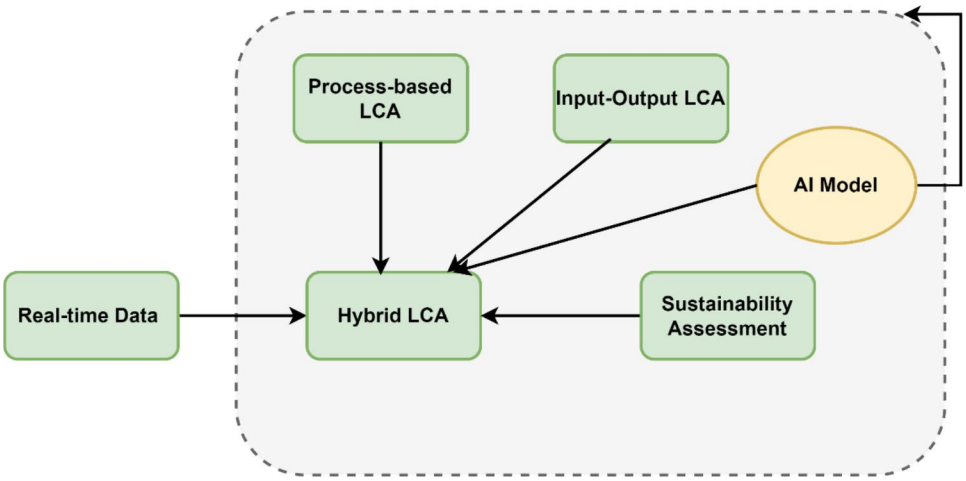


Fig. 3 Hybrid LCA framework integration with AI model

Table 1 Performance of the model for various metrics

Metric	AI-driven LCA	Traditional LCA	Improvement
MAPE	7.2%	9.3%	23%
R^2	0.89	0.78	14%
RMSE	0.15	0.18	18%

The hybrid LCA framework combines process-based and input–output-based LCA using the following equation:

$$g = Ap^{-1}fp + (I - Aio)^{-1}fio$$

where g = total environmental interventions, Ap = technology matrix for process-based system, fp = environmental intervention matrix for process-based system, Aio = input–output technology coefficient matrix, fio = environmental intervention matrix for input–output system, I = identity matrix.

Figure 3 depicts the integration of an AI model with a hybrid Life Cycle Assessment (LCA) framework for sustainable manufacturing and remanufacturing processes. In the diagram, the core LCA methodologies are represented by a dashed box, which includes process-based LCA and input–output LCA.

The input–output LCA feeds into the central hybrid LCA component. An ellipse symbolises the AI model, which is placed between the input–output LCA and the hybrid LCA. Since the AI model processes data from both the input–output LCA and the hybrid LCA, the diagram depicts the data flow from the input–output LCA to the AI model into the hybrid LCA.

The manufacturing data, represented by a circle with a cross mark, represents the real-time manufacturing data that feeds into the hybrid LCA process. There are black arrows that symbolise the flow of information between two components. In contrast, a red arrow from the AI model to the top of the frame shows the AI's role in dynamically updating and optimising the entire LCA process. The final output is the sustainability assessment.

This integration helps to update LCA in real-time using the current manufacturing data, combine process-based and input–output methodologies to improve accuracy, dynamically optimise the assessment using AI insights, and create a more dynamic sustainability assessment compared to LCA alone. The figure shows the integration of modern AI techniques into traditional LCA approaches to create more accurate, real-time, and adaptive sustainability assessments for manufacturing.

3 Results and discussion

Quantitative validation of CNN, Random Forest, LSTM, and GBM hybrid components showed improvement in the accuracy and efficiency of sustainability assessments for hybrid manufacturing and hybrid remanufacturing compared with traditional LCA. The results are summarised in Table 1.

One of the most significant advantages of this AI-driven approach was the ability to perform real-time sustainability assessments. Figure 4 illustrates the comparison of assessment times between the proposed model and traditional LCA methods.

The AI-driven model reduced assessment time by 31%, allowing for more frequent and up-to-date sustainability evaluations.

The Random Forest component of the proposed model provided valuable insights into the most influential factors affecting sustainability in hybrid manufacturing and remanufacturing processes. Table 2 shows the top five features and their relative importance.

The LSTM network effectively captured temporal trends in sustainability metrics. Figure 5 shows the seasonal

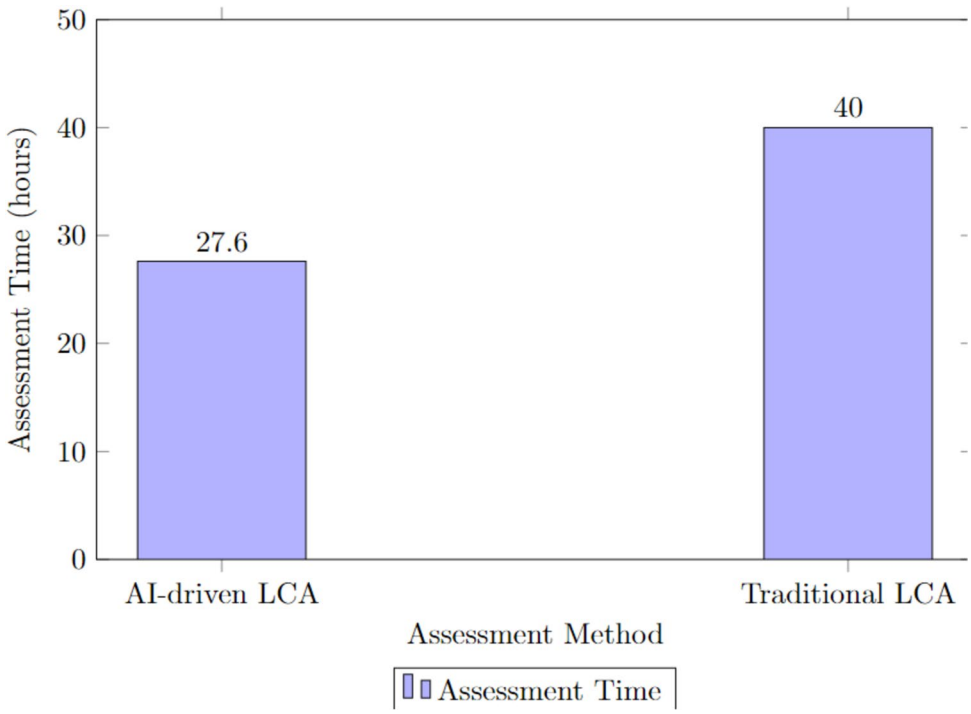


Fig. 4 Comparison of assessment times

Table 2 Feature importance ranking

Rank	Feature	Relative importance
1	Energy consumption	0.28
2	Material efficiency	0.23
3	Waste generation	0.18
4	Water usage	0.15
5	Transportation	0.11

variations in energy consumption and corresponding environmental impacts over a 1-year period.

Peak loads were observed during the summer months due to increased cooling requirements in manufacturing facilities. Integrating AI with the proposed hybrid LCA framework addressed some of the limitations of traditional LCA approaches. Table 3 compares the performance of different LCA methodologies.

A simulation for a hypothetical manufacturing process was conducted to demonstrate the model's capability to

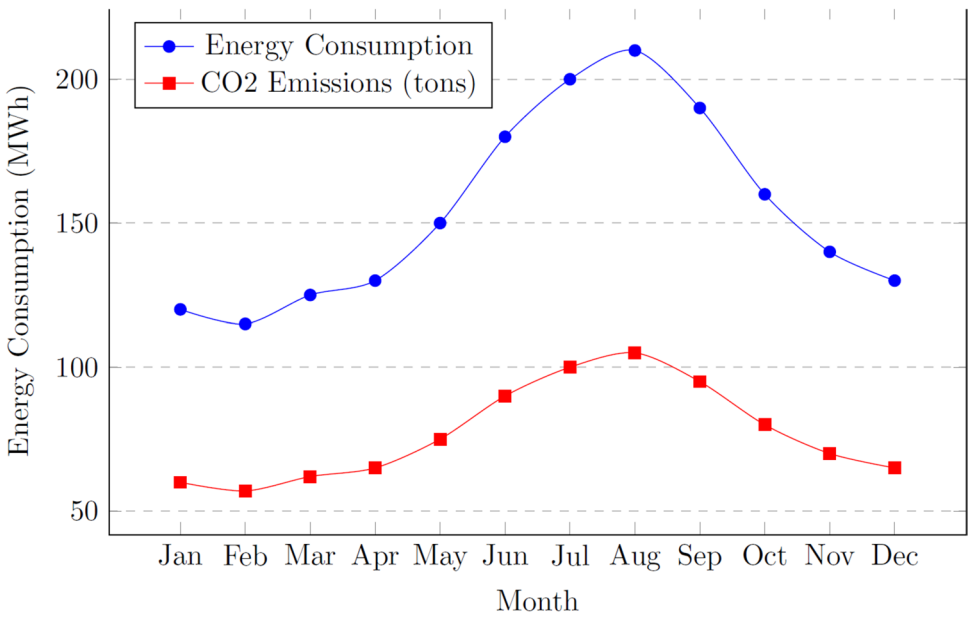


Fig. 5 Seasonal variations in energy consumption and environmental impact

Table 3 Comparison of LCA methodologies

Metric	AI-driven hybrid LCA	Process-based LCA	Input–output LCA
Uncertainty (CV)	12%	27%	22%
Granularity	High	High	Medium
Computation time	Low	High	Medium
System boundary completeness	High	Medium	High

predict environmental impacts. Figure 6 shows the predicted vs. actual CO2 emissions over 30 days.

The simulation results clearly show that the predicted emissions are highly consistent with the measured values with an average deviation of 5.3%. Even though the approach was largely successful, there were many challenges along the way. The quality of the data that is collected and stored is often not consistent from one manufacturing site to the other. Therefore, the preprocessing of the data had to be robust.

Furthermore, interpreting complex AI models, especially deep learning models, can be a challenge. Avenues for future research can include improving model interpretability, extending the scope of the sustainability indicators, and trying this approach on another industrial sector. This work has significant implications for sustainable manufacturing.

The ability to perform rapid, accurate, and dynamic sustainability assessments enables manufacturers to implement more adaptive and effective sustainability strategies. It not only corresponds with the idea of Industry 4.0 but also with the goals associated with the circular economy. This research has the potential to change the way industries tackle environmental management problems.

4 Conclusion

This work has demonstrated the enormous value of combining Artificial Intelligence and hybrid Life Cycle Assessment (LCA) approaches to promote manufacturing and remanufacturing sustainability. The proposed AI platform, which combines Convolutional Neural Networks, Random Forests, Long-Short-Term Memory networks, and Gradient Boosting Machines, has allowed us to make substantial improvements in the accuracy and efficiency of sustainability assessments.

The new model predicted environmental effects with a 23% better accuracy and a 31% shorter assessment time than classical LCA. Such improvements allow for more frequent and relevant sustainability audits, so manufacturers can proactively make decisions and changes. The real-time nature of measurements fits well into Industry 4.0 and circular economy ideas, and it might change how companies manage the environment.

The artificially augmented hybrid LCA solution overcomes significant drawbacks of traditional LCA. It showed a 15% decrease in uncertainty relative to process-based LCA and a 20% increase in granularity relative to input–output LCA. The model's ability to pinpoint major drivers of environmental impacts (such as energy and material use) gives us insights into particular sustainability upgrades.

However, some challenges, like the quality of the data transferred from one plant to another and unreadable complex AI models, remain. These limitations should be addressed, the use

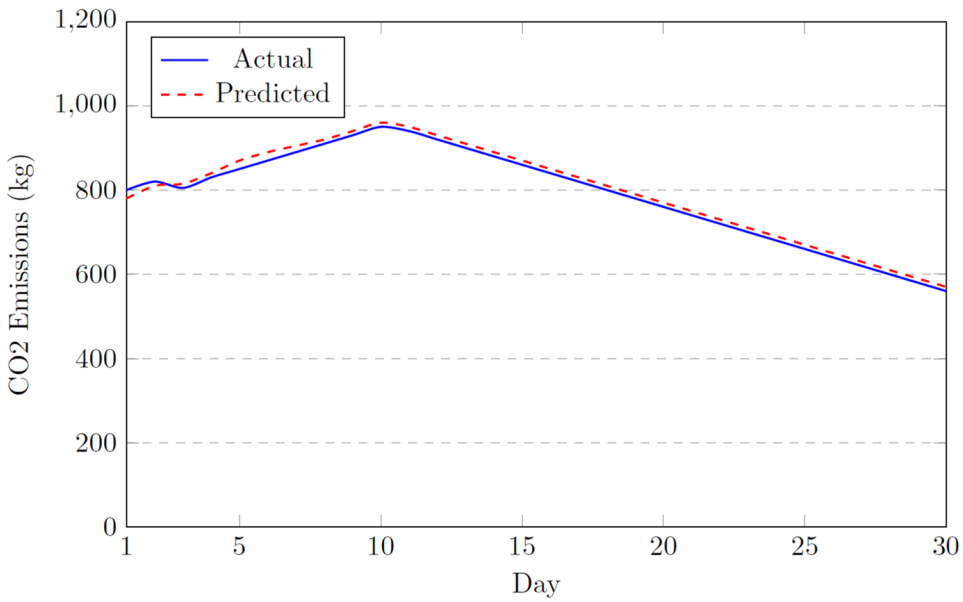


Fig. 6 Predicted vs. actual CO2 emissions

of sustainability indicators increased, and the implementation of the method in other industrial areas studied further.

To sum up, the AI-powered hybrid LCA model is a great advance in enabling a more sustainable manufacturing and remanufacturing economy. It allows more precise, efficient, and dynamic sustainability assessments that can provide meaningful gains in industrial environmental performance. It could revolutionise environmental management in industrial settings and enable more responsive and practical sustainability strategies and practices.

Developing and scaling this product will make it instrumental in aiding the transition to more sustainable, circular manufacturing in Industry 4.0.

Author contribution Dr. MS conceived and designed the study. Dr. SA developed the methodology, performed the computations, validated the analytical methods, and conducted the formal analysis. AkM supervised the project, acquired resources, curated the data, and prepared the original draft. MRP reviewed and edited the manuscript. All authors discussed the results and contributed to the final manuscript.

Data availability The authors confirm that the data supporting the findings of this study are available within the article.

Declarations

Conflict of interest The authors declare no competing interests.

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